CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 5: **Adolescence: A Stage of Growth and Change** | Quick revision, one page

Adolescence and puberty

- Adolescence: the stage between childhood and adulthood, roughly 11 to 19 years (the teenage years).
- Puberty: the stage when sexual (reproductive) maturity starts; begins earlier in girls than in boys.
- Changes: rapid growth in height, change in body shape, voice change, growth of body and facial hair, oil and sweat glands more active (pimples).

Hormones and glands

- Hormones: chemical messengers made by endocrine glands (no ducts) and carried by the blood.
- Pituitary: master gland (growth hormone). Thyroid: thyroxine. Pancreas: insulin.
- Testes: testosterone (male). Ovaries: oestrogen (female).

Changes in boys and girls

- Boys: voice box enlarges (Adam's apple), voice deepens, facial hair, broader shoulders, sperms produced.
- Girls: breasts develop, hips widen, ovaries release eggs, menstruation begins (menstrual cycle about 28 to 30 days).

Care during adolescence

- Balanced diet (proteins, calcium, iron), regular exercise, enough sleep, cleanliness and bathing.
- Mood swings and emotional changes are normal; talk to parents, teachers or friends.

MASTERMIND